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Winter Chilling in Areas with Mild Winters: Its Measurement and Supplementation

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Keywords: *Actinidia deliciosa* var. *deliciosa*, *Prunus persica*, *Vitis vinifera*

Abstract

Early attempts to measure winter chilling, using the Utah Chill Unit model, were unsuccessful as negative totals were obtained at Pietermaritzburg, despite the fact that low chill peach trees produce well. This was due to high daytime temperatures (25 to 30°C) negating the beneficial effects of low night temperatures. Positive Utah Chill Units (PCU) can be accumulated, by ignoring negative daily totals. These PCU have been used to produce maps of South Africa showing how much winter chilling accumulates each month in different areas. Both the Israeli dynamic model and positive chill units have been shown to be equally accurate in estimating the amount of winter chilling in both cold and mild areas of South Africa. Production of low-chill 'Allison' kiwifruit at Pietermaritzburg has been improved (trebled) by evaporative cooling in winter, especially under 30 to 40% artificial shade. Dormex® (hydrogen cyanamide) sprays in late winter have had a synergistic effect with evaporative cooling in giving the greatest production. By intermittent overhead sprinkling, when daytime temperatures exceed 16°C in winter, wet bud temperatures have been kept below 19°C, with a beneficial effect on winter chilling. Evaporative cooling should start when daytime temperature exceeds 16 to 18°C provided the night temperature had been below 12°C. Ideally, sprinkling cycles should be long enough to wet all the buds effectively, and the OFF time should be short enough to prevent bud temperatures rising too high. Production of early table grapes has been improved (improved bud break and increased number of bunches produced) in the lower Orange river area of South Africa, in some years. Intermittent sprinkling is used on a large scale as insurance in case of inadequate winter chilling in mild winters.

INTRODUCTION

All deciduous fruit plants require chilling during the winter rest period to enable their buds to sprout evenly, and for good flowering and fruit set to occur in spring. Insufficient winter chilling results in prolonged rest or delayed foliation, and consequently poor production and quality of fruits. While the major deciduous fruit producing areas in South Africa are in the SW and S Cape, colder areas of the Free State and lower Orange river, and there is limited production in subtropical KwaZulu/Natal where problems with lack of winter chilling often occur.

In the subtropics (mean temperature of coldest month between 13 and 18°C, and occurrence of light frost) the winter day/night temperature variations are much greater than in temperate regions, due to typically cloudless days and nights, and greater radiative heat loss at night. High day temperatures (>16 and especially >19°C) have an adverse effect on chilling accumulation even when night temperatures are sufficiently cold (Richardson et al., 1974; Erez et al., 1993; Allan and Burnett, 1995). This paper summarizes some 20 years of research on methods of evaluating winter chilling and its supplementation.

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WINTER CHILLING MODELS (Allan, 1999)

The old method of counting the number of "chilling hours below 7°C" during the winter months was found to be inaccurate under South African conditions (Linsley-Noakes et al., 1994). This method has been replaced by Utah chill units described below.

Utah Chill Units

The Utah chill unit (UCU) model (Richardson et al., 1974) was developed in Utah (40°N, high altitude), USA. It is based on allocating 0, 0.5 or 1 unit per hour at specified temperatures and negative units above 16°C. This model was adopted by the deciduous fruit industry in the South Western Cape region of South Africa (Matthee, 1982). However, further research (Linsley-Noakes et al., 1994) showed that this model is not accurate in areas with high winter day temperatures (over 20°C). The high negative totals accumulated during warm spells (20 to 25+ °C) led to variable results and negative UCU totals being recorded in some years at Pietermaritzburg, even though there was enough chilling for low-chill peaches (*Prunus persica*) to sprout and flower (Blaine and Allan, 1979; Allan and Burnett, 1995; Allan et al., 1997).

Dynamic Model

This model, developed in Israel (Fishman et al., 1987), is based on the hypothesis that the level of dormancy completion depends on the level of a certain dormancy breaking factor, which is accumulated by a two-step process (Fig. 1). The first step is assumed to be a dynamic, reversible process of formation and destruction of a thermally labile "intermediate" product, formed from a precursor (chilling temperatures promote the reaction while high temperatures favour the reversal of the reaction). When a critical level (1.0) or quantum of the "intermediate" is amassed, it is transferred, irreversibly, in the second step to a quantum (one portion) of the stable dormancy breaking factor [now called Chilling Portion (CP) (A. Erez, 1998, pers. commun.). Once a Chilling Portion has been formed it cannot be broken down by subsequent high temperatures.

Using the mathematical formulae for the Dynamic model (A. Erez, 1997, pers. commun.), the effects of different temperature regimes on the variations in accumulation of the "intermediate" to the critical level (1.0), and hence Chilling Portion (CP) formation, were calculated, and are shown in Fig. 2 (Allan et al., 1999). At constant 6°C it can be seen that it takes 28 h for the level of the "intermediate" to build up to the critical level of 1.0, when one CP is fixed, irreversibly.

At constant 12°C, it takes much longer (40 h) for one CP to be fixed, while at constant 15 or 25°C the critical level is never reached and the level of "intermediate" reaches a plateau at 0.8 or 0.29 respectively. The higher the temperature the lower the plateau of the "intermediate". Unless the level of "intermediate" reaches the critical level of 1.0, there is no formation of a Chilling Portion, and hence no accumulation of chilling. However, a moderate temperature of 15°C for 8 h, followed by 6°C, results in a faster initial build up of "intermediate" and the critical level is reached only after 26 h.

Conversely, constant 6°C for 24 h, followed by 25°C, resulted in the level of "intermediate" falling 50%, from 0.9 to 0.44, in 3 h (i.e. 3 h at 25°C cancelled the beneficial effect of the previous 14 h at 6°C). The level of "intermediate" continued to decrease more slowly and, after 11 h, reached the plateau of 0.29 (Fig. 2). This clearly demonstrates the adverse effect of a few hours of high day temperatures on negation of chilling accumulation, whereas, after prolonged high temperatures, there is no further adverse effect.

In Israel, the Dynamic model has been adopted as their measure of winter chilling, and the numbers of Chilling Portions, accumulated in all the major fruit growing areas, are calculated and made available regularly to growers by their meteorological service (A. Erez, 1997, pers. commun.).

Daily Positive Chill Units

Linsley-Noakes et al. (1995) suggested that, because high temperature negation affects only the current level of the "intermediate" product, and the fact that Chilling Portions, once formed, are fixed irreversibly, long periods of high temperature will have a limited negating effect on chilling. He thus suggested deleting the carry-over effect of negating temperatures from one day to the next in the UCU model. If the 24 hour total is negative it is counted as zero, but, if the daily total is positive, it is added to the seasonal total, thus giving daily Positive Chill Units (PCU). These values are now calculated from weather data in South Africa by the Institute for Soil, Climate and Water, Pretoria.

Approximate PCU

Linsley-Noakes developed tables of approximate PCUs, based on the daily maximum and minimum temperatures recorded in a Stevenson screen (Linsley-Noakes et al., 1995) (Table 1). These approximate PCU can serve as a useful guide although the results tend to be lower than those obtained by hourly calculation (Allan and Burnett, 1995).

Best Models

Linsley-Noakes et al. (1994) showed that the amount of winter chilling in different areas of the SW and S Cape could be estimated with similar degrees of accuracy by both the Dynamic model (CP) and the Positive Chill Unit model (PCU), and both are suitable for areas with mild winters as well as cold winters. In contrast the Utah chill unit model is suitable only for areas with cold winters (Erez et al., 1990; Linsley-Noakes et al., 1994).

Comparison of the results obtained by both the Dynamic and PCU models shows that the ratio of CP:PCU is slightly under 1:20 (Allan et al., 1999). Thus the units of the two models can be compared, approximately, using this ratio. Thus a peach cultivar with a fairly high chilling requirement of 800 PCU requires about 40 CP.

Maps of Chilling Accumulation

The daily positive chill unit (PCU) model was used, using data from 325 temperature stations in South Africa and neighbouring states, to develop very useful maps of South Africa showing the amount of winter chilling that can be expected each month, in average years (Schulze et al., 1997).

EVAPORATIVE COOLING

Kiwifruit

Kiwifruit (*Actinidia deliciosa* var. *deliciosa*) has been grown experimentally at Pietermaritzburg (29° 40' S, 30° 25' E, altitude 700 m) since 1970's but it was not until the advent of hydrogen cyanamide (Dormex®) in the 1980's that improved and more uniform bud break was achieved and limited production was obtained (Brunette, 1992). Subsequently it was shown that evaporative cooling (EC), by intermittent overhead sprinkling during the heat of the day, improved bud break and yield, especially with the synergistic effect of Dormex® and evaporative cooling (Allan et al., 1994; 1997).

Evaporatively cooled kiwifruit buds were generally kept from rising to temperatures above 18°C during warm winter days, provided the buds were kept wet. However where buds dried out their temperatures rose within a few minutes to the level of un-sprinkled buds. Erez (1995) stated that whenever night temperatures reach 12°C or lower, chilling will accumulate if day temperature is kept below 19°C.

After the unusually warm winter in 1998, with about one third of the chilling of the previous two years, even evaporatively cooled vines had reduced reproductive bud break (19 vs. 28-31%) but control (non evaporatively cooled) vines (under 30% shade) produced very few reproductive shoots (4%). These control plants produced an average of only 6 fruits per vine that season, compared to several hundred fruits on the plants in the evaporative cooling treatment (Allan et al., 1999).

Grapes

Nir et al. (1988) showed that overhead micro-sprinklers decreased the temperatures of exposed and shaded grape (*Vitis vinifera*) buds, in the Jordan Valley in Israel, from 30 and 25 to 16 and 13°C respectively. However with ON and OFF periods of 1.5 and 15 to 20 minutes respectively, bud temperatures rose about 8°C, to around 22°C, during the OFF period.

In the Blouputs valley of the lower Orange river area of South Africa (lowest recorded mean monthly temperature of 13.4°C in July), where early table grapes are produced two weeks before the Upington area, production of 'Thompson Seedless' grapes was reported to be much lower than at Upington in certain years (P. du Plessis, pers. commun., 1996).

In May 1996 a 0.5 ha experiment was started on evaporative cooling of 'Thompson Seedless' grapes in the Blouputs valley. The results were promising with earlier and improved bud break and an 18% increase in the number of bunches produced during the early, high price period in November (Allan and Hattingh, 1998). However, both in 1996 and in 1997, the winters were unusually cold, and yields of Control plants were also fairly good.

In contrast, the winter of 1998 was exceptionally mild, and the response to evaporative cooling was outstanding. An average of eight bunches of grapes were produced per cane under evaporative cooling compared with three bunches per cane in the control plot. Hydrogen cyanamide (Dormex®) is applied routinely, early in July, by the grape growers, to promote good bud break. The combination of evaporative cooling and Dormex raised bud break to over 80%, compared to 50% for Dormex® alone, and 0% without any treatment.

As a consequence of the good results achieved in 1998, evaporative cooling has been applied by growers on a large scale (circa 200 ha) since then. While a good response is not obtained every year, evaporative cooling is used as a form of insurance in case of insufficient chilling during May and June.

Evaporative cooling should start when day temperatures rise above 16 to 18°C, provided night temperatures were below 12°C. Ideally, sprinkling cycles should be long enough to wet all the buds effectively, and the OFF time should be short enough to prevent bud temperatures rising too high.

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Tables

Table 1. Daily positive Utah chill unit (PCU) estimation table determined on the assumption that diurnal temperatures follow a sine curve for the heating cycle, followed by a linear cooling cycle (Linsley-Noakes et al., 1995).

MAXIMUM TEMPERATURE °C

MINIMUM TEMP °C	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
-5	8	6.5	6.5	5	5	4	2.5	2	2.5	0	0	0	0	0	0	0	0	0	0
-4	12	11	11	10	10	11	9.5	8	8	6	4.5	4.5	1.5	0.5	0	0	0	0	0
-3	13	12	12	11	11	11	10	9	8.5	7	5	3.5	2.5	1.5	0	0	0	0	0
-2	14	13	13	12	12	12	11	8.5	8.5	7	5	5	4.5	2	2	0	0	0	0
-1	15	15	14	14	14	14	12	10	8.5	8.5	6	5.5	3	1.5	1	0	0	0	0
0	15	15	15	14	14	14	12	12	9.5	9.5	7	6.5	3.5	1.5	1.5	0	0	0	0
1	17	17	17	16	16	16	14	13	12	11	8	7	4	2.5	1.5	0.5	0.5	0	0
2	21	20	20	19	19	19	17	15	13	12	10	8.5	6	4	3	2	1	1	0
3	22	22	22	21	21	20	18	16	15	13	12	9.5	7.5	5	4	3	1.5	1.5	0.5
4	22	22	21	21	21	19	18	16	15	13	11	9.5	6.5	4	2.5	2.5	1	0	0
5	22	22	21	20	20	19	17	15	14	13	10	8	5	3.5	2.5	2	0.5	0	0
6	23	22	21	19	19	19	16	15	13	12	10	8	4.5	2.5	2	0	0	0	0
7	21	21	20	19	19	18	15	14	12	11	8.5	6.5	3	1.5	0	0	0	0	0
8	20	20	20	19	19	16	13	12	11	10	6.5	6	1.5	0.5	0	0	0	0	0
9	20	18	17	15	15	14	11	9	8	6.5	4	2.5	0	0	0	0	0	0	0
10	20	17	12	12	12	12	9	7.5	6.5	5	2	0.5	0	0	0	0	0	0	0
11	19	15	12	12	12	12	8.5	6.5	5.5	4.5	1.5	0.5	0	0	0	0	0	0	0
12	19	15	12	12	12	12	7.5	4	3.5	2	0	0	0	0	0	0	0	0	0

Figures

Chilling temps-->

Precursor **Intermediate ----> Chilling Portion**

<—High temps **(Level varies)** **(Fixed irreversibly)**

Fig. 1. Dynamic model, a two step process in formation of chilling portions (Dormancy breaking factor) (Fishman et al., 1987).

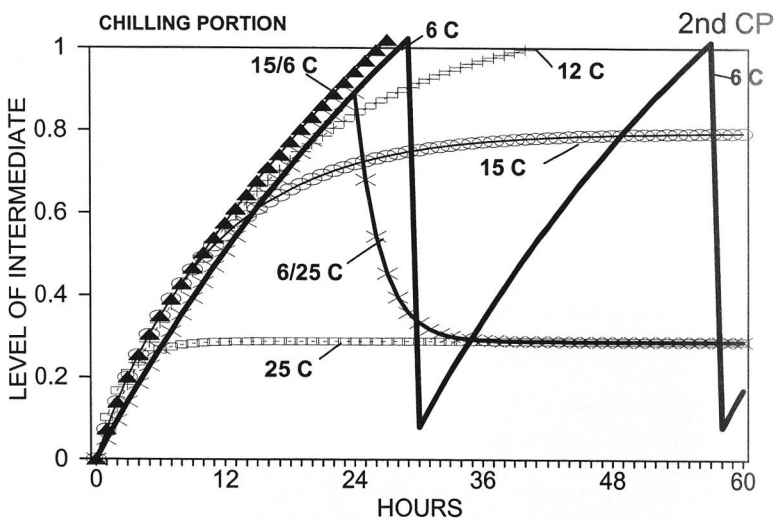


Fig. 2. The effects of different temperature regimes on the accumulation of the "intermediate" and Chilling Portions, according to the Dynamic model (from Allan et al., 1999).